Caption text for the PENK presentation slides

The AlphaFold render is a **visual element**, and that is a legitimate use. The only hard rule is that the caption must not claim something the model does not show. Below is wording that is both presentable and defensible.

Suggested order

1. **Native enkephalins** — what the natural gene makes: 6 Met-, 1 Leu-enkephalin
2. **Domain architecture** — what the engineered molecule is, where the cargo sits
3. **AlphaFold confidence** — folded anchor vs disordered cargo tail
4. **Processing** — the KR cuts releasing the octapeptide
5. **Receptor complex** — the released peptide bound, with the 12.05 Å callout

The narrative runs: floppy carrier → proteolytic release → bound conformation that only exists after excision. That sequence is why the disorder in shot 2 is a feature of the story rather than a weakness to explain away.

Slide: the native enkephalins (`enkephalin_met_leu_overview.png`)

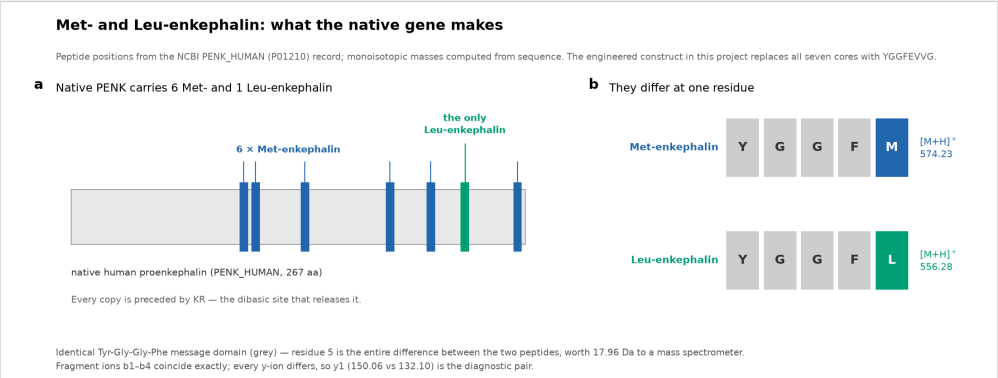


Figure 1 — Shot 1. Native proenkephalin peptide inventory and the single-residue difference between the two enkephalins.

The opening shot — what the natural gene makes, before the engineered construct is introduced.

Say this:

> Human proenkephalin is 267 residues and carries 6 copies of Met-enkephalin and exactly one Leu-enkephalin. The two peptides share the same Tyr-Gly-Gly-Phe message domain and differ only at residue 5.

Verified values on this slide:

quantity	value
native proenkephalin length	267 aa (NCBI PENK_HUMAN, P01210)
Met-enkephalin copies	6, at residues 100, 107, 136, 186, 210, 261
Leu-enkephalin copies	1, at residue 230
Met-enkephalin [M+H]+	574.23 Da
Leu-enkephalin [M+H]+	556.28 Da
mass difference	17.96 Da
diagnostic fragment pair	y1 = 150.06 (Met) vs 132.10 (Leu)

The mass-spectrometry point worth making on screen: b1–b4 are **identical** between the two peptides because the first four residues are the same; every y-ion differs. So the y-series, not the b-series, tells them apart.

Do not say: "the engineered construct makes Met- and Leu-enkephalin." It does not — it replaces all seven cores with `YGGFEVVG` and contains neither native core (verified: 0 copies of YGGFM, 0 of YGGFL). The native peptides are context for the design, not its product.

Leu-enkephalin structure: there is currently **no** structural model for Leu-enkephalin in this project. A co-fold of both peptides against both the delta- and mu-opioid receptors is queued on the cluster (job 17958370, `pgpu`, estimated start 2026-08-15). Until it lands, do not show a Leu-enkephalin structure — the only way to produce one now would be to mutate the Met model's side chain, which is not an independent prediction and must not be presented as one.

Slide: domain architecture (`slide_penk_domain_architecture.png`)

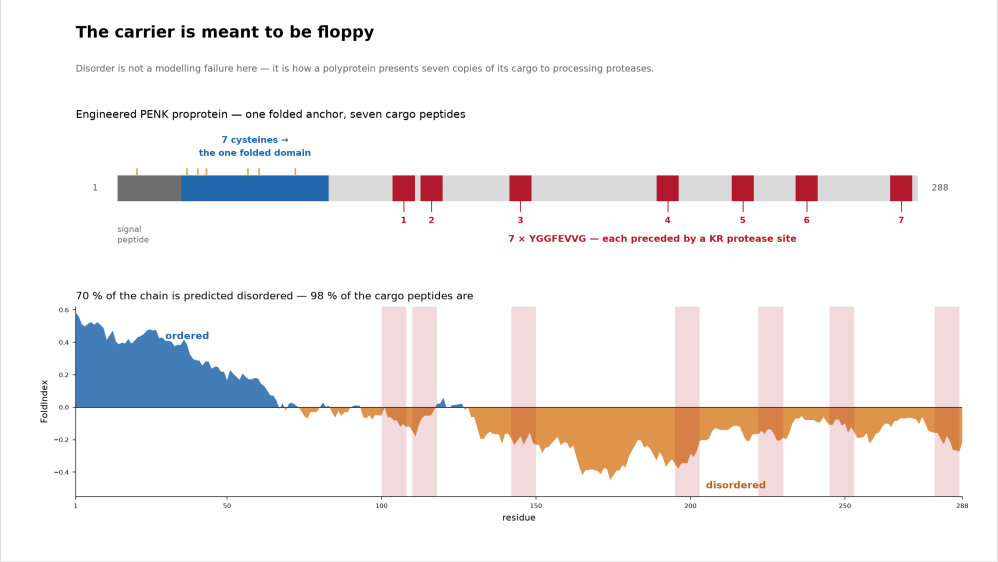


Figure 2 — Shot 2. Engineered proprotein domain architecture over the FoldIndex disorder trace.

16:9, 2666 × 1500 px, drops straight onto a slide.

Talk track: "One folded domain, held by seven cysteines. Everything after it is deliberately floppy — and that is where the seven copies of the therapeutic peptide live, each one preceded by the KR site the cell's proteases cut."

> Wording note: say **preceded by KR**, not "flanked by KR or KK." All seven cores have KR immediately upstream, but the downstream pair varies (KR, KK, RG, RR, RF). "Flanked by KR or KK" overstates what the sequence shows — and the same correction applies to the native gene.

This slide carries the argument the AlphaFold render cannot: where the cargo is and why the disorder is functional. Show it immediately before or after the render.

Verified values on this slide: 7 × `YGGFEVVG` at residues 100, 110, 142, 195, 222, 245, 279; cysteines at 8, 26, 30, 33, 48, 52, 65; FoldIndex 70.5 % of the chain disordered, 98.2 % of the 56 core residues disordered.

Slide: the AlphaFold proprotein render

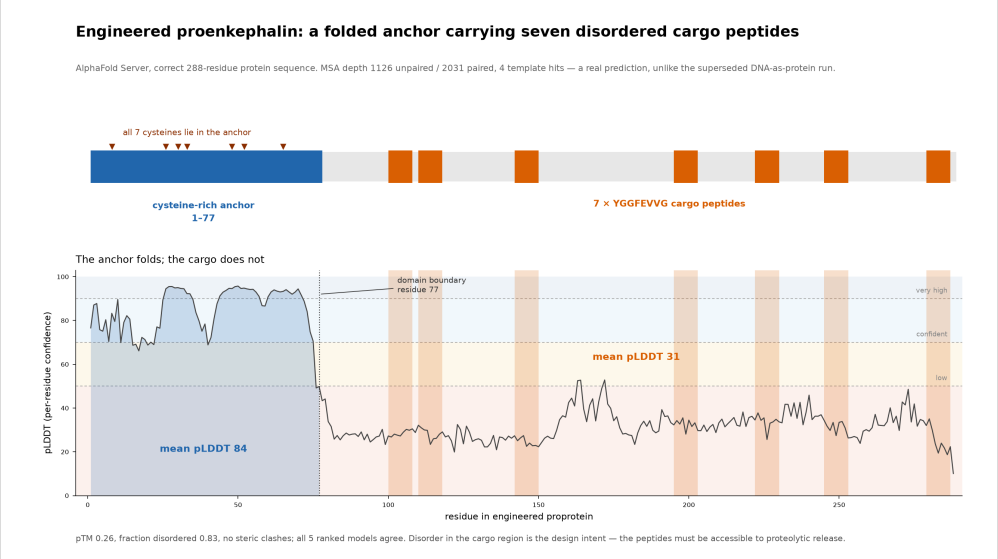


Figure 3 — Shot 3. AlphaFold per-residue confidence: folded anchor (residues 1–77, mean pLDDT 84) against the disordered cargo tail (mean pLDDT 31).

Status: the corrected run is complete. Values below are read from its own output files and are cleared for use on screen. The earlier run — in which the 885-nt DNA cloning sequence was submitted in the protein field — is **superseded and must not be shown**; A, C, G and T are all valid amino-acid letters, so it returned a confident-looking model of a molecule that does not exist.

Say this:

> AlphaFold model of the engineered PENK proprotein (288 aa). A folded, cysteine-rich anchor at residues 1–77 carries a long disordered tail holding the seven enkephalin copies — which is how the carrier presents them to processing proteases.

Do not say:

- "AlphaFold structure of the engineered enkephalin" — the render is the **carrier**, 288 residues. The enkephalin is 8 of them, ×7. Calling the whole thing "the enkephalin" is the mistake the audience will catch.
- "predicted binding conformation" — there is no receptor in this model.
- "the model is low confidence, so it failed" — a low global pTM is the expected signature of a mostly-disordered carrier, not a bad prediction. The anchor is confidently folded; say so.

On-screen numbers — verified, safe to display:

quantity	value	source
chain length	288 aa	submitted sequence, identical to the stored construct
ordered anchor	residues 1–77, mean pLDDT **84**	per-residue pLDDT, model 0
cargo tail	mean pLDDT **31**	per-residue pLDDT, model 0
cysteines in anchor	**7 of 7**	sequence positions 8, 26, 30, 33, 48, 52, 65
pTM	**0.26**	`summary_confidences_0.json`
fraction disordered	**0.83**	`summary_confidences_0.json`
steric clashes	none, all 5 ranked models	`summary_confidences_*.json`
MSA depth	1126 unpaired / 2031 paired, 4 templates	run's own `msas/` and `templates/`

Keep the two disorder methods distinct. AlphaFold's own figure is `fraction_disordered` = **0.83**. The FoldIndex figure is **70.5 %** of the chain — a different calculation. Cite whichever you use by name; never present one as the other.

Assets: `slide_engPENK_alphafold_confidence.png` (16:9 domain bar over the pLDDT trace), `engPENK_alphafold_per_residue_plddt.csv`, `engPENK_alphafold_CORRECTED_model_0.cif`.

Slide: the excised peptide in the receptor

Assets: `DOR_engPENK_complex_annotated.pdb`, plus the ChimeraX / PyMOL scripts.

Say this:

> The processed octapeptide bound in the δ-opioid receptor. Boltz-2 co-folded model — a different predictor from AlphaFold.

Do not say: "AlphaFold confidence" over this render. The colours are Boltz-2 pLDDT, and they describe a **receptor-bound** peptide, not a free one. The palette matches AlphaFold's so the shots cut together visually; the caption must still name the right method.

The number worth putting on screen: the Tyr1-OH → Phe4-ring pharmacophore distance, **12.05 Å engineered vs 8.05 Å native** — measured from the coordinates, and a real visual difference between the two molecules.